MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL,
FLIP-FLOPS, MONOLITHIC SILICON

This amendment forms a part of Military Specification MIL-M-38510/[illegible]A, dated 1 May 1979, and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

* TABLE II, delete and substitute as follows:

"TABLE II. Electrical test requirements.

MIL-STD-883 test requirement	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7, 8,10,11	1,2,3,7, 8
Group A test requirements (method 5005)	1,2,3,7,8, 9,10,11	1,2,3,7, 8
Group B test requirements (method 5005) subgroup 5	1,2,3,9, 10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,7,3
Additional electrical subgroups for group C periodic inspections	N/A	10,11
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.3c)."

PAGE 20

* FIGURE 6: Delete and substitute new figure as printed on page 3 of this amendment.

PAGE 21

* FIGURE 7: Delete and substitute new figure as printed on page 4 of this amendment.

PAGE 24

* TABLE III, Terminal conditions (pins not designated may be high $\geq$ 2.0 V or low $\leq$ 0.7 V or open): Delete "0.7 V" and substitute: "0.8 V". Make this same change on pages 25 and 26.

PAGE 27

* TABLE III, Group A, device type 02: Delete and substitute new table III as printed on page 5 of this amendment.

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(SEE NOTE

6.7, delete and substitute as follows:

"6.7 Manufacturers' designations. Manufacturers' circuits included in this specification are designated as shown in table IV.

TABLE IV. Manufacturers' designations.

Device type	National Semiconductor	Texas Instruments	Signetics	Motorola	Fairchild
	Circuits				
	A	B	C	D	E
01	X	X	X	X	X
02	X	X	X	X	X

"6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes."

The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Air Force - 17

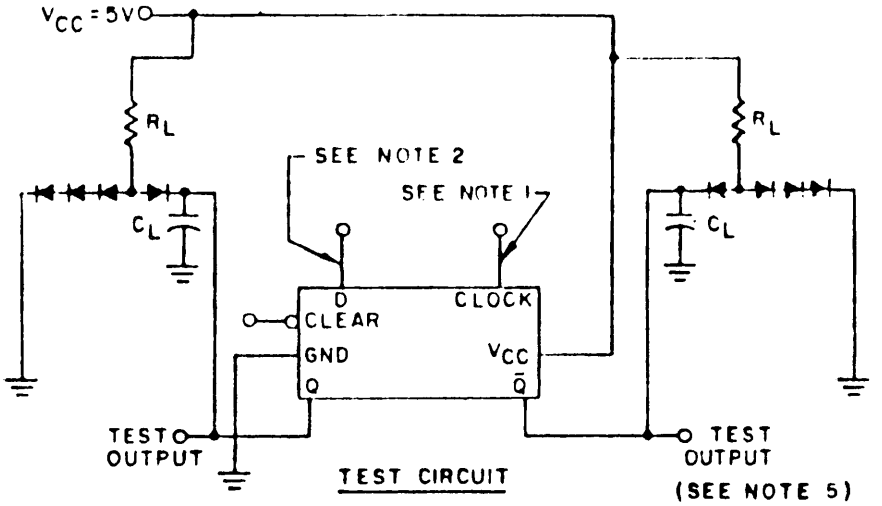
Review activities:
Air Force - 11, 19, 85, 99
DLA - ES

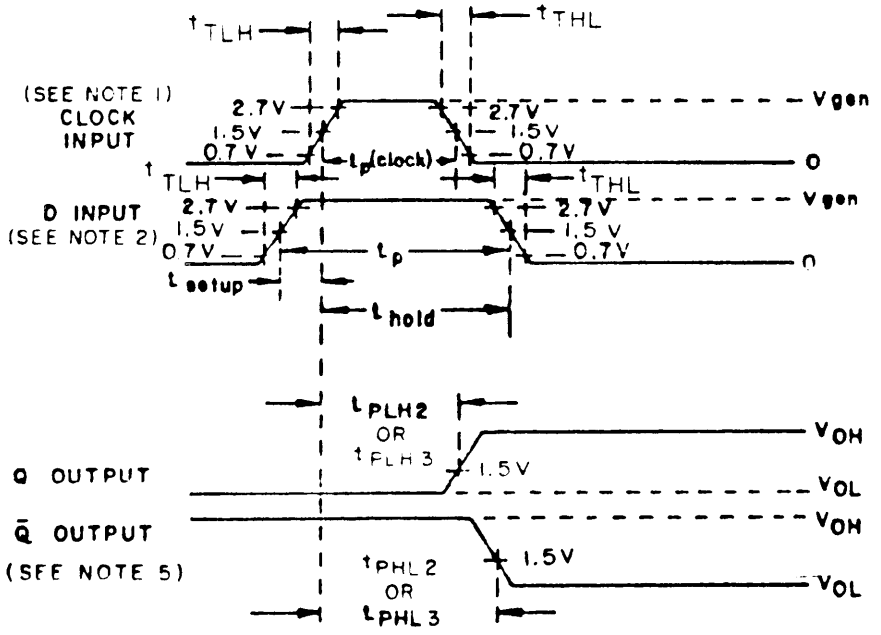
Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-F671)

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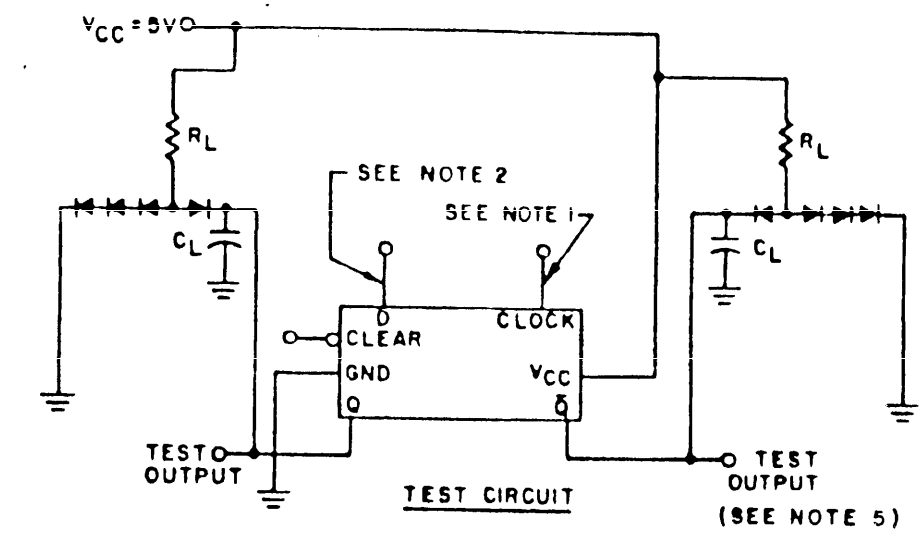


VOLTAGE WAVEFORMS

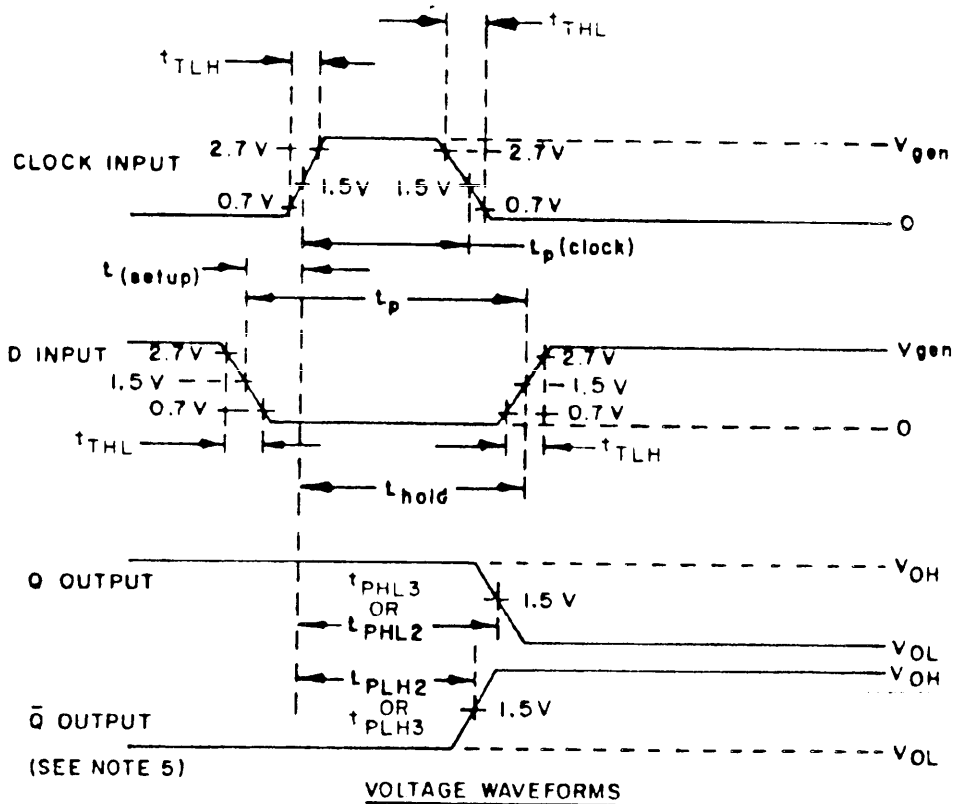
NOTES:
1. Clock input has the following characteristics: V_{gen} = 3 V minimum, t_p = 20 ns, $t_{THL} = t_{TLH} \leq 10$ ns, and PRR $\leq$ 1 MHz. When testing F_{MAX} PRR = 25 MHz.
2. D input pulse has the following characteristics: V_{gen} = 3 V minimum, $t_{TLH} = t_{THL} \leq 10$ ns, t_p = 30 ns, t_{SETUP} = 25 ns, t_{HOLD} = 5 ns and PRR $\leq$ 0.5 MHz. When testing F_{MAX}, PRR = 12.5 MHz at 50% ±15% duty cycle.
3. R_L = 390Ω ± 5%; C_L = 50 pF ± 10% (including jig and probe capacitance).
4. All diodes are 1N3064, or equivalent.
5. $\bar{Q}$ output applies to device type 02 only.

FIGURE 6. Synchronous switching test circuit (high-level data) for device types 01 and 02.

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TEST CIRCUIT



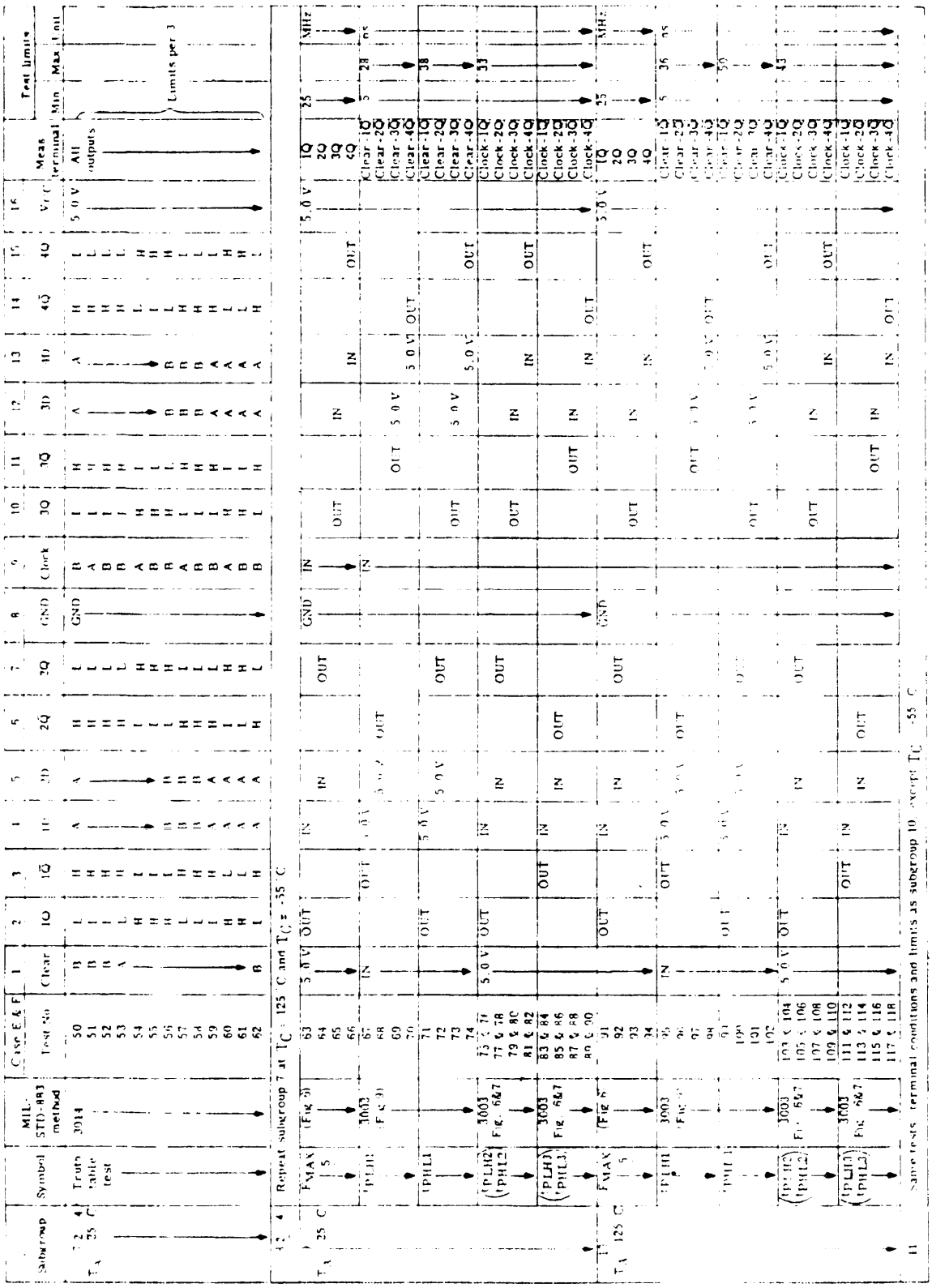
VOLTAGE WAVEFORMS

NOTES:
1. Clock input has the following characteristics: V_{gen} = 3 V minimum, t_p = 20 ns, $t_{THL} = t_{TLH} \leq 10$ ns, and PRR $\leq$ 1 MHz. When testing F_{MAX} PRR = 25 MHz.
2. D input pulse has the following characteristics: V_{gen} = 3 V minimum, $t_{TLH} = t_{THL} \leq 10$ ns. t_p = 30 ns, t_{SETUP} = 25ns, t_{HOLD} = 5 ns and PRR $\leq$ 0.5 MHz. When testing F_{MAX}, PRR = 12.5 MHz at 50% ±15% duty cycle
3. R_L = 390Ω ± 5% ; C_L = 50 pF ± 10% (including jig and probe capacitance).
4. All diodes are IN3064, or equivalent.
5. $\bar{Q}$ output applies to device type 02 only.

FIGURE 7. Synchronous switching test circuit (low-level data) for device types 01 and 02.

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TABLE III. Group A inspection for device type 02. Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).



NOTES:
1. H: Minimum (GND, then V_{CC}).
2. Tests as in subgroups 7 and 8 shall be performed in the sequence specified.
3. Output voltages shall be either:
   a. H: 2.4 V minimum and L: 0.4 V maximum when using a high-speed checker-double comparator, or
   b. H: 1.5 V and L: 1.5 V when using a high-speed checker-single comparator.
4. Only a summary of attributes data is required.
5. F_{MAX}: minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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